Rum — OSA Digital Twin: System Architecture

A synthetic EHR risk profile and real wearable-style physiological signals are fused into one clinical flag.

STATIC (patient history)

Synthea

Open-source synthetic patient generator — fully synthetic EHR, zero real patient data.

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Weighted comorbidity score

Rule-weighted: obesity, hypertension, prediabetes, type 2 diabetes, existing sleep apnea diagnosis.

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Static risk (cohort percentile, 0–1)

Patient's rank within the synthetic population's weighted risk score.

DYNAMIC (real-time signal)

MIT-BIH Polysomnographic Database

PhysioNet, open-access. Real ECG + nasal airflow + SpO2, expert-scored apnea/hypopnea annotations.

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30-second windows → HRV + breathing features

Heart rate, RR-interval SDNN, LF/HF ratio, breath rate, breath-interval variability, breath amplitude.

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XGBoost + causal smoothing

Leave-one-patient-out validated. Trailing 3-min average → dynamic risk score (0–1).

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Fusion

risk = 0.3 × static risk + 0.7 × dynamic risk (hand-set weights, illustrative)

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Threshold flag

LOW (<0.50) **MODERATE** (0.50-0.65) **HIGH** (≥0.65)

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Clinician Dashboard (concept)

Patients ranked by fused risk — overnight risk timeline — EHR risk tags. Not a diagnostic device.